

Chapter 05

Costing Techniques

Standard Costing

Standard Costing involves setting predetermined standard costs for each unit of product manufactured for Direct Material, Direct Labour, and Overhead. This serves as a benchmark for cost control, performance evaluation, and pricing.

Components of Standard Cost

- **Standard Direct Material Cost:** $\text{Standard Quantity (SQ)} * \text{Standard Price (SP)}$
- **Standard Direct Labour Cost:** $\text{Standard Hours (SH)} * \text{Standard Rate (SR)}$
- **Standard Overhead Cost:** $\text{Budgeted Overhead} / \text{Standard Activity Level}$

Variance Analysis

The difference between the Actual Cost (AC) and the Standard Cost (SC) is the Cost Variance. Analysis helps pinpoint inefficiencies.

Material Price Variance (MPV) = Actual Quantity (AQ)(SP – AP)

Material Usage/Quantity Variance (MQV) = SP*(SQ - AQ)

- **A favorable variance** occurs when $AC < SC$
- **An unfavorable variance** occurs when $AC > SC$

Exercise

A company's standard material cost is 2 kg @ \$10/kg. Actual production used 420 kg at \$9.50/kg for an output that should have used 400 kg.

Variance	Formula	Calculation	Result
MPV	$AQ * (SP - AP)$	$\$420 * (\$10 - \$9.50)$	\$210 (F)
MQV	$SP * (SQ - AQ)$	$\$10 * (400 \text{ kg} - 420 \text{ kg})$	\$200 (U)
Total Material Variance	$MPV + MQV$	$\$210 \text{ F} - \200 U	\$10 (F)

By-Products and Joint Products

- **Joint Products:** Two or more products of significant sales value produced simultaneously from a common raw material and process.
- **Joint Costs:** Costs incurred up to the split-off point where joint products become separately identifiable. **Physical Measurement Method, Market Value at Split-off Method, Net Realizable Value (NRV) Method, etc.**
- **By-Products:** Products of minor sales value produced incidentally from the main process. **Cost Reduction Method, Net Realizable Value (NRV) Method, etc.**

Allocation of Joint Costs (Joint Products)

Physical Measure Method: Allocates costs based on a physical unit (like weight or volume) of each product at the split-off point.

$$\text{Allocated Joint Cost} = \text{Total Joint Cost} \times \frac{\text{Physical Units of Product}}{\text{Total Physical Units of All Products}}$$

Sales Value at Split-off Method: Allocates costs based on the relative sales value of each product at the split-off point. This is often preferred because it links costs to revenue-generating ability.

$$\text{Allocated Joint Cost} = \text{Total Joint Cost} \times \frac{\text{Sales Value of Product at Split-off}}{\text{Total Sales Value of All Products at Split-off}}$$

Net Realizable Value (NRV) Method: This is used if further processing is needed after the split-off point.

NRV=Final Sales Value–Separable Costs

This NRV is then used as the basis for allocation, similar to the Sales Value method.

Problem 1: Physical Measurement Method

A company processes raw material into two joint products.

Product	Output (kg)
A	3,000
B	2,000

Total joint cost = **Tk. 250,000.**

Allocate joint costs using the **physical unit method.**

Problem 2: Sales Value at Split-Off Method

Three joint products emerge at the split-off point:

Product	Quantity	Split-off Price
X	2,000 kg	Tk. 40 per kg
Y	1,200 kg	Tk. 55 per kg
Z	1,800 kg	Tk. 30 per kg

Joint cost = **Tk. 400,000.**

Allocate joint cost using the **market value at split-off** method.

Problem 3: Net Realizable Value (NRV) Method

Two products require further processing:

Product	Output	Selling Price	Further Cost
A	5,000 units	Tk. 20/unit	Tk. 25,000
B	3,000 units	Tk. 30/unit	Tk. 15,000

Total joint cost = **Tk. 300,000.**

Calculate:

1. NRV of each product
2. Allocation of joint cost using NRV method

Accounting for By-Products

- Production Method: By-product revenue is treated as a reduction in the Joint Cost of the main products.
- Sales/Other Income Method: By-product revenue is treated as Other Income on the Income Statement at the time of sale.

Foreign Exchange Exposure Management: Risks

- 1. Transaction Risk:** The risk that currency fluctuations will affect the cash flows associated with an existing foreign-currency-denominated transaction (e.g., an outstanding receivable or payable).
- 2. Translation Risk:** The risk that currency fluctuations will affect the value of a foreign subsidiary's assets and liabilities when their financial statements are consolidated into the parent company's home currency.
- 3. Economic Risk (Operating Risk):** The risk that a company's future cash flows and market value will be affected by unexpected currency fluctuations over the long term.

Strategies to Reduce Risk (Hedging)

Internal Strategies:

- **Invoicing in Home Currency:** Shifting the transaction risk to the foreign party.
- **Netting:** Offsetting foreign currency payables against receivables in the same currency.
- **Leading and Lagging:** Paying/receiving funds *earlier* (leading) or *later* (lagging) based on currency expectation.

External Strategies (Financial Instruments):

- **Forward Contracts:** A binding agreement to buy or sell a fixed amount of foreign currency at a specified rate on a future date. This *locks in* the exchange rate.
- **Currency Options:** Gives the holder the *right*, but not the obligation, to buy (call) or sell (put) a currency at a specified rate (strike price) on or before a future date.

Types of Budgets

Budgets are formal financial plans detailing expected revenues and costs.

1. **Master Budget:** A comprehensive set of budgets that covers all aspects of an organization's planned activities for a specified period. Includes:
 1. **Operating Budgets:** Sales, Production (Material, Labour, Overhead), Selling & Admin.
 2. **Financial Budgets:** Cash, Capital Expenditure, Budgeted Balance Sheet.
2. **Fixed (Static) Budget:** Prepared for a **single, planned level of activity**. Not useful for performance evaluation if actual activity differs significantly from the planned level.
3. **Flexible Budget:** Adjusted to show expected costs and revenues at **various activity levels**. Excellent for performance evaluation and cost control by comparing actual results to the budget *adjusted* for the actual output level. Flexible budgets take into account how changes in activity affect costs.

A flexible budget is an estimate of what revenues and costs should have been, given the actual level of activity for the period. When a flexible budget is used in performance evaluation, actual costs are compared to what the costs should have been for the actual level of activity during the period rather than to the static planning budget. This is a very important distinction. If adjustments for the level of activity are not made, it is very difficult to interpret discrepancies between budgeted and actual costs.

(A flexible budget adjusts to show what costs should be for the actual level of activity.)

Budgeting and Control in Tanneries and Leather Products

- **Sales Budget:** Forecast of sales volume and revenue by product line (e.g., finished leather, shoes, bags). This drives the production plan.
- **Production Budget:** Determines the required units of finished leather/products to meet sales demand and target inventory levels.

Units to Produce = Budgeted Sales + Desired Closing Stock - Opening Stock

- **Material Budget (Hides, Chemicals):** Forecasts the required quantity and cost of raw materials, heavily dependent on the quality and yield rate of the hides/skins.
- **Labour Budget:** Estimates direct labour hours and cost, considering the skilled nature of work and potential for piece-rate structures.
- **Overhead Budget:** Allocation of costs like power, water treatment, and facility expenses. **Budgetary Control** involves comparing actual costs against budgeted amounts and taking corrective action.

Exercise: Budget Schedules

Mynor Corporation manufactures and sells a seasonal product that has peak sales in the third quarter. The following information concerns operations for Year 2—the coming year—and for the first two quarters of Year 3:

a. The company's single product sells for \$8 per unit. Budgeted unit sales for the next six quarters are as follows (all sales are on credit):

	Year 2 Quarter				Year 3 Quarter	
	1	2	3	4	1	2
Budgeted unit sales	40,000	60,000	100,000	50,000	70,000	80,000

Exercise: Budget Schedules (Cont'd)

b. Sales are collected in the following pattern: 75% in the quarter the sales are made, and the remaining 25% in the following quarter. On January 1, Year 2, the company's balance sheet showed \$65,000 in accounts receivable, all of which will be collected in the first quarter of the year. Bad debts are negligible and can be ignored.

c. The company desires an ending finished goods inventory at the end of each quarter equal to 30% of the budgeted unit sales for the next quarter. On December 31, Year 1, the company had 12,000 units on hand.

Exercise: Budget Schedules (Cont'd)

d. Five pounds of raw materials are required to complete one unit of product. The company requires ending raw materials inventory at the end of each quarter equal to 10% of the following quarter's production needs. On December 31, Year 1, the company had 23,000 pounds of raw materials on hand.

e. The raw material costs \$0.80 per pound. Raw material purchases are paid for in the following pattern: 60% paid in the quarter the purchases are made, and the remaining 40% paid in the following quarter. On January 1, Year 2, the company's balance sheet showed \$81,500 in accounts payable for raw material purchases, all of which will be paid for in the first quarter of the year.

Requirement:

Prepare the following budgets and schedules for the year, showing both quarterly and total figures:

1. A sales budget and a schedule of expected cash collections.
2. A production budget.
3. A direct materials budget and a schedule of expected cash payments for purchases of materials.

Exercise: Variance Analysis Using a Flexible Budget

Harrald's Fish House is a family-owned restaurant that specializes in Scandinavian-style seafood. Data concerning the restaurant's monthly revenues and costs appear below (q refers to the number of meals served):

	Formula
Revenue	$\$16.50q$
Cost of ingredients	$\$6.25q$
Wages and salaries	$\$10,400$
Utilities	$\$800 + \$0.20q$
Rent	$\$2,200$
Miscellaneous	$\$600 + \$0.80q$

Requirements:

1. Prepare the restaurant's planning budget for April assuming that 1,800 meals are served.
2. Assume that 1,700 meals were actually served in April. Prepare a flexible budget for this level of activity.
3. The actual results for April appear below. Prepare a flexible budget performance report for the restaurant for April.

Revenue	\$27,920
Cost of ingredients	\$11,110
Wages and salaries	\$10,130
Utilities	\$1,080
Rent	\$2,200
Miscellaneous	\$2,240